

NOTEBOOK

OVERVIEW

ZJU-CHINA2026





Notebook — Protein Hydrogel Scaffold & Ligation System

Timeline of the Work

Date (2026)	Work performed	Record
6 July	Recovery of the glycerol stocks of the scaffold protein (2KY-5) and the Ligation Syste proteins (SnoopTag2-sfGFP, SnoopLigase2, SUMO-DogTag) in DH5α; transformation of the plasmids into BL21(DE3)	1
7 July	Colony PCR confirmation of the 2KY-5 cloning strain	2
8–9 July	Small-scale expression of 2KY-5 and the three Ligation Syste proteins; gradient induction trials	3
10 July	Whole-cell boiling lysis and SDS-PAGE to verify successful induction	4
12–14 July	Small-scale Ni-NTA affinity purification of the three Ligation Syste proteins	5
14 July	Western blot of the purified Ligation Syste proteins with an anti-His primary antibody; confirmation of purification and of the optimal induction concentration	6
16–18 July	ITC purification of 2KY-5; intermediate samples taken throughout for SDS-PAGE	7
19–21 July and 2–4 August	Iterative optimisation of the 2KY-5 purification conditions (salt and temperature gradients), monitoring by SDS-PAGE; the best parameters were fixed for later reuse	8
22–24 July	Dialysis and desalting of purified 2KY-5, followed by lyophilisation	9
26–28 July	Large-scale expression of the three Ligation Syste proteins	10
29–30 July	Large-scale Ni-NTA purification of the Ligation Syste proteins and dialysis of the purified proteins	11
31 July	Ligation reaction of the Ligation Syste proteins; SDS-PAGE evaluation of the ligation result and of the coupling efficiency	12
8–9 August	Recovery and culture of the 2KY-5/DH5α cloning strain; plasmid mini-prep	13
10–13 August	Troubleshooting of the low plasmid yield: plasmid concentration and adjustment of the culture and extraction conditions; optimised parameters established (chloramphenicol added at mid-exponential phase, heated incubation during elution)	14
16–19 August	Increase of the 2KY-5 repeat number by PRe-RDL to give the new plasmid 2KY-10; transformation of the cloning strain DH5α	15
20–21 August	Plasmid mini-prep of the 2KY-10/DH5α cloning strain and transformation into the expression strain BL21(DE3)	16
22–25 August	Induction of 2KY-10 expression; whole-cell boiling lysis and SDS-PAGE verification of induction	17
26–29 August	ITC purification of 2KY-10, dialysis, desalting and lyophilisation	18
1 September	Preparation of the Ru/SPS photo-crosslinking system (2/20 mM) from the lyophilised protein powder and crosslinking under blue light	19
3 September	Repeat of the photo-crosslinking of the hydrogel backbone with adjusted blue-light wavelength parameters; optimised crosslinking result obtained	20
4–8 September	Recovery, transformation, production and purification of SnoopTag2-2KY-5-SnoopTag2, and construction, transformation, production and purification of SnoopTag2-2KY-10-SnoopTag2 in the same way	21
10 September	Ligation reaction of SnoopTag2-2KY-5-SnoopTag2 and SnoopTag2-2KY-10-SnoopTag2 with SUMO-DogTag and SnoopLigase2; SDS-PAGE verification of the ligation	22
12–14 September	Extraction of the prepared hydrogel in culture medium for 24 h; CCK-8 cytotoxicity test of the extract	23
19–21	Cell adhesion experiment on the prepared hydrogel	24

September		

Experiment Records

1 Recovery of glycerol stocks and plasmid transformation

Date: 6 July 2026

Procedures

1. The glycerol stocks of E. coli DH5α carrying the scaffold plasmid 2KY-5 and the Ligation Syste plasmids SnoopTag2-sfGFP, SnoopLigase2 and SUMO-DogTag were taken from −80 °C storage, streaked onto LB agar plates containing the appropriate antibiotic and incubated at 37 °C overnight for recovery.
2. A single colony of each strain was inoculated into 5 mL of LB broth with antibiotic and grown at 37 °C, 200 rpm to mid-exponential phase; the cultures were used to prepare fresh glycerol stocks and to verify the plasmids.
3. Chemically competent BL21(DE3) cells were thawed on ice for 5–10 min; 50 µL aliquots were mixed with 1–2 µL (about 50–100 ng) of each of the four plasmids by gentle pipetting, and a negative control without DNA was set up in parallel.
4. The mixtures were kept on ice for 30 min, heat-shocked at 42 °C for 45–90 s and returned to ice for 2 min.
5. 800 µL–1 mL of pre-warmed LB broth without antibiotic was added and the cells were recovered at 37 °C, 200 rpm for 45–60 min.
6. 50–200 µL of each recovery culture was spread onto LB agar with the appropriate antibiotic; the plates were left upright at 37 °C for 10–15 min and then inverted and incubated for 12–16 h.
7. Single colonies were picked for colony PCR verification (record 4.2) or for inoculation of liquid cultures.

Note

All four plasmids gave well-separated single colonies on the selective plates, while the negative control remained clean, indicating that the glycerol stocks were viable and that the transformation of BL21(DE3) was successful for every construct. Plates were stored at 4 °C and used within one week for the expression trials.

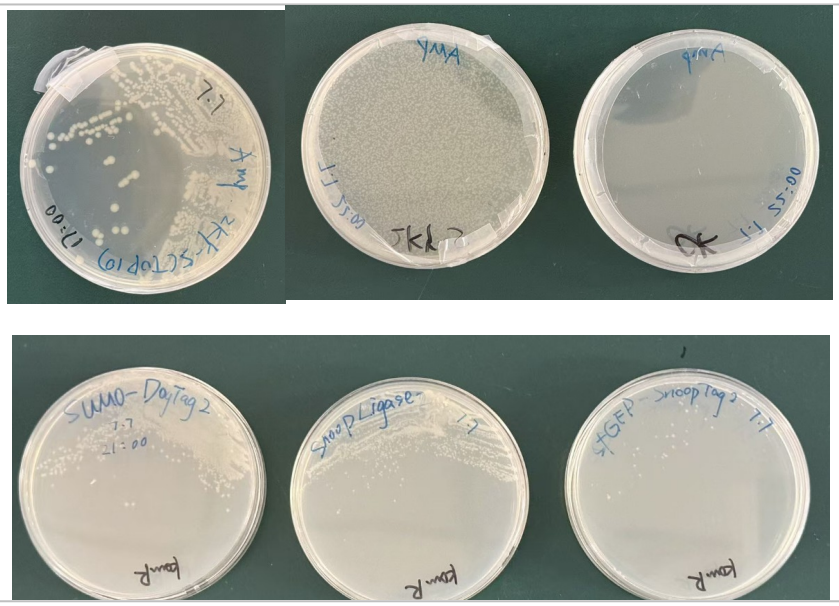


Figure 1. Colony morphology of the four transformed BL21(DE3) strains on selective LB agar plates (2KY-5, SnoopTag2-sfGFP, SnoopLigase2, SUMO-DogTag).

2 Colony PCR verification of the 2KY-5 cloning strain

Date: 7 July 2026

Procedures

1. Eight well-isolated colonies of the 2KY-5/DH5α strain were marked on the plate and each was transferred with a sterile tip into 10 µL of nuclease-free water and resuspended by pipetting.
2. A 25 µL colony PCR reaction was set up for each clone: 12.5 µL of 2× Taq master mix, 0.5 µL each of the forward and reverse primer (10 µM) and 1–2 µL of the colony suspension, made up with nuclease-free water. One primer annealed to the vector backbone and the other to the insert, so that the parental plasmid alone could not give a false positive.
3. A positive control (purified 2KY-5 plasmid), a no-template control and an empty-vector control were included in the same run.
4. The programme used was: initial denaturation at 94 °C for 3 min; 30 cycles of 94 °C for 30 s, 55 °C for 30 s and 68 °C for 1 min per kb; final extension at 68 °C for 5 min; hold at 4 °C.
5. 5–10 µL of each reaction was analysed on a 1% (w/v) agarose gel against a DNA marker, and the band sizes were compared with the expected insert size.
6. Positive clones were inoculated into 5 mL of LB broth with antibiotic and grown at 37 °C, 200 rpm overnight for glycerol-stock preparation and sequencing.

Note

Colonies showing a single band of the expected size were scored as positive; the number of positive clones out of the eight screened is recorded in the figure below. The positive clones were retained as the working stock of the 2KY-5 cloning strain for the subsequent plasmid preparation and for the PRe-RDL construction.

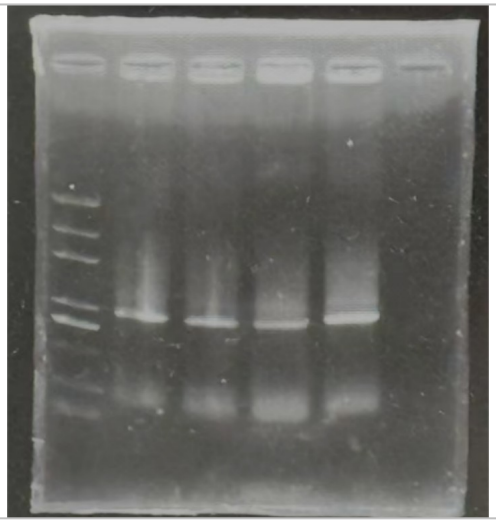


Figure 2. Agarose gel of the colony PCR products of the eight 2KY-5/DH5α clones (DNA marker; L2-5:1–8, clones; −, no-template control).

3 Small-scale expression and gradient induction trials

Date: 8–9 July 2026

Procedures

1. Starter cultures of the four expression strains (2KY-5, SnoopTag2-sfGFP, SnoopLigase2 and SUMO-DogTag in BL21(DE3)) were prepared in 10 mL of LB broth with antibiotic and grown at 37 °C, 200 rpm for 16 h.
2. Each starter culture was diluted 1:100 into 50 mL of fresh LB broth with antibiotic and grown at 37 °C, 200 rpm until the OD600 reached 0.5–0.8.
3. Each culture was divided into parallel flasks and induced with a gradient of IPTG (0.1, 0.25, 0.5 and 1.0 mM) so that the optimal inducer concentration could be identified for every construct.
4. Different induction temperatures and durations were tested on the same set of cultures, including 16 °C for 20 h, 25 °C for 8 h, 30 °C for 4 h and 37 °C for 4 h.
5. One millilitre of culture was taken immediately before induction and at the end of the induction for every condition; the cells were normalised by OD600, pelleted and resuspended in 1× SDS loading buffer for later SDS-PAGE analysis.
6. The remaining culture volumes were harvested by centrifugation at 4 °C, 4,000–6,000 g for 15 min, washed once with pre-chilled PBS and stored at −80 °C.

Note

All four constructs were expressed, but the soluble yield depended strongly on the induction condition. Low-temperature induction (16–25 °C with a longer duration) gave the best balance between yield and solubility for the SELP backbone 2KY-5, whereas the three Ligation Syste proteins tolerated shorter induction at 25–30 °C. The combination that gave the strongest band in the whole-cell samples was carried forward to the SDS-PAGE verification and used for all later preparations.

4 SDS-PAGE verification of induction by whole-cell boiling lysis

Date: 10 July 2026

Procedures

1. The pre-induction and post-induction cell samples collected during the gradient trials were thawed, normalised to the same OD600 and pelleted.
2. Each pellet was resuspended in 20–30 µL of 1× SDS loading buffer and heated at 95 °C for 5–10 min to lyse the cells and denature the proteins (whole-cell boiling lysis).
3. Samples were centrifuged briefly and 5–15 µL of each supernatant was loaded onto a 12% (w/v) polyacrylamide gel, with a pre-stained molecular-weight marker in the first lane and the pre-induction samples next to their induced counterparts.
4. Electrophoresis was run at a constant 180 V for about 45–60 min until the bromophenol blue front reached the bottom of the gel.
5. The gel was stained with Coomassie brilliant blue for 20–60 min and destained with deionised water until the background was clean.
6. The gel was imaged and the band intensities were compared between the lanes to confirm induction and to rank the induction conditions.

Note

For all four constructs a band of the expected molecular weight was clearly stronger in the induced lanes than in the corresponding pre-induction lanes, confirming that induction was successful. The 2KY-5 band appeared at the expected position for the low-repeat SELP backbone, and the Ligation Syste proteins gave bands at their respective expected sizes. The condition giving the strongest specific band was fixed as the standard induction condition and reused in every later expression.

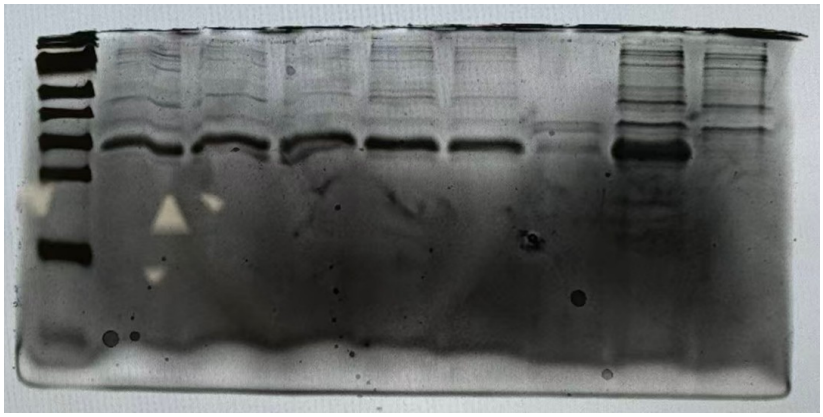


Figure 4. SDS-PAGE of the whole-cell boiling-lysate samples before and after induction for 2KY-5 and the three Ligation Syste proteins (L1:marker; L2-6:CL-sup;L7:CL-pel; L8: after induction;L9:before induction).

5 Small-scale Ni-NTA affinity purification of the Ligation system proteins

Date: 12–14 July 2026

Procedures

1. The frozen cell pellets of SnoopTag2-sfGFP, SnoopLigase2 and SUMO-DogTag were resuspended at 1:10 (w/v) in ice-cold Ni-NTA buffer (50 mM Tris-HCl pH 8.0, 300 mM NaCl) supplemented with 10 mM imidazole, EDTA-free protease inhibitor cocktail and 1 mM PMSF.
2. The suspensions were lysed on ice by sonication (30–40% power, 5 s on / 10 s off, 10–20 min in total) until they cleared.
3. The lysates were centrifuged at 4 °C, 30,000–35,000 g for 30–40 min and the supernatants were collected as clarified lysates; a sample of each pellet and supernatant was retained for SDS-PAGE.
4. Each clarified lysate was incubated with Ni-NTA agarose resin (1–2 mL of resin slurry per litre of culture) at 4 °C with gentle inversion for 1 h.
5. The resin/lysate slurry was poured into a gravity column and drained; the flow-through was collected for SDS-PAGE to assess the binding efficiency.
6. The resin was washed with 3 × 10 column volumes of Ni-NTA buffer containing 10 mM imidazole until the A280 of the flow-through returned to baseline.
7. Bound protein was eluted with Ni-NTA buffer containing 200 mM imidazole; 0.5–1 mL fractions were collected and the protein peak was followed by A280.
8. Selected fractions were analysed by SDS-PAGE and the fractions containing the target protein were pooled.
9. The pooled eluates were concentrated and exchanged into PBS or Tris-borate buffer pH 7.4 by ultrafiltration and dialysis, and the protein concentration was determined by A280.

Note

All three Ligation Syste proteins were obtained in the elution fractions, with the target band dominating the Coomassie-stained gel; the flow-through and wash fractions contained only minor amounts of the target protein, indicating efficient binding. SnoopLigase2 was kept at a concentration below 300 µM during handling to limit aggregation. The purified proteins were aliquoted and stored at −80 °C for the ligation test.

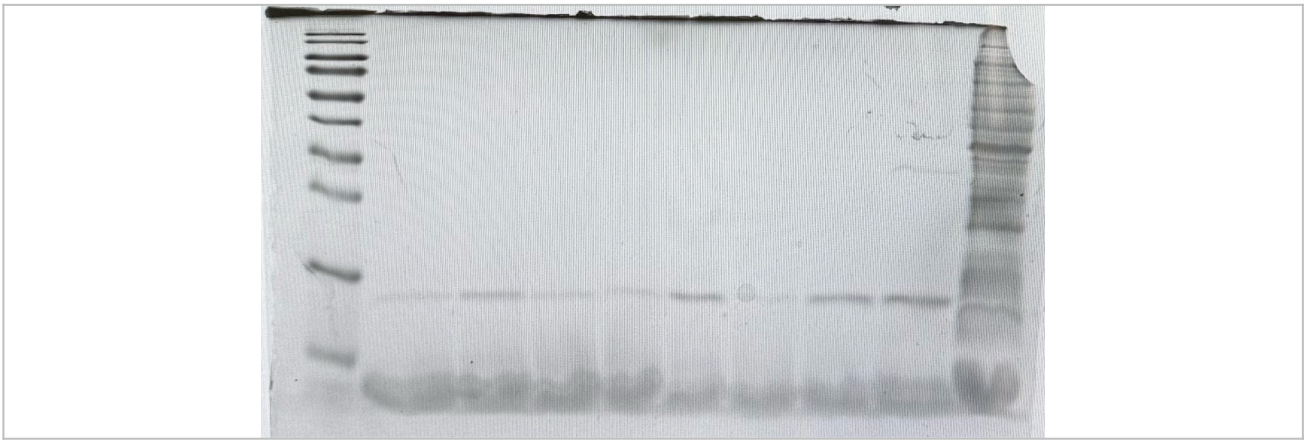


Figure 5. Small-scale Ni-NTA purification of the three Ligation Syste proteins: SDS-PAGE of the lysate, pellet, flow-through, wash and elution fractions.

6 Western blot verification with an anti-His antibody

Date: 14 July 2026

Procedures

1. The purified Ligation Syste proteins and the corresponding whole-cell samples were separated by SDS-PAGE on a 12–16% (w/v) gel as described in record 4.4.
2. The proteins were transferred onto a nitrocellulose membrane by wet transfer at 100 V for 60–90 min in an ice bath; the transfer efficiency was checked by Ponceau S staining.
3. The membrane was blocked in PBST containing 5% (w/v) skimmed milk at room temperature for 1 h.
4. The membrane was incubated with an anti-His primary antibody diluted 1:1000–1:5000 in blocking solution at 4 °C overnight with gentle agitation.
5. The membrane was washed 3–4 times with PBST for 5–10 min each.
6. An HRP-conjugated secondary antibody diluted 1:5000–1:10000 was applied and incubated at room temperature for 1 h with agitation.
7. The membrane was washed 4 × 5 min with PBST and once with PBS, and the signal was developed with ECL substrate and imaged on a chemiluminescence imager.
8. Band intensities were quantified with image-analysis software.

Note

The anti-His antibody detected single specific bands at the expected molecular weights for SnoopTag2-sfGFP, SnoopLigase2 and SUMO-DogTag, confirming both the identity of the purified proteins and that the Ni-NTA purification was feasible. The signal intensities of the induced samples were used to compare the induction conditions, and the IPTG concentration giving the strongest specific signal with the least background was selected as the optimal induction concentration for the large-scale expression.

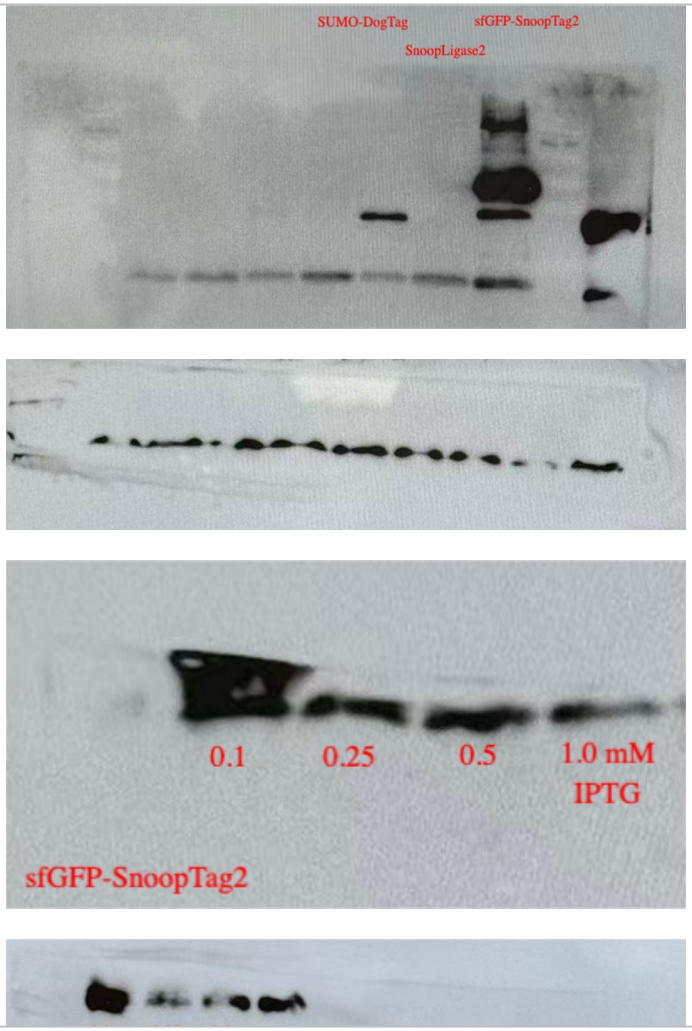


Figure 6. Western blot of the purified Ligation Syste proteins probed with an anti-His primary antibody .

7 ITC purification of the 2KY-5 scaffold protein

Date: 16–18 July 2026

Procedures

1. The 2KY-5 cell pellet was resuspended at 1:10 (w/v) in ice-cold PBS supplemented with protease inhibitors immediately before use.
2. The suspension was lysed on ice by high-pressure homogenisation or sonication until it cleared.
3. The lysate was centrifuged at 4 °C, 15,000–20,000 g for 20–30 min and the clarified supernatant was retained.
4. First transition (hot precipitation): NaCl was added to a final concentration of 1.5–3 M, the solution was mixed and incubated at 37 °C for 10–30 min until it became turbid.
5. The suspension was centrifuged at 30–37 °C, 15,000–20,000 g for 10–20 min; the supernatant containing the contaminants was discarded and the aggregated target protein pellet was kept.
6. Cold dissolution: the pellet was resuspended in ice-cold PBS and incubated on ice for 10–30 min until the protein had redissolved.

7. The solution was centrifuged at 4 °C, 15,000–20,000 g for 10–20 min and the supernatant containing the target protein was kept.
8. The hot-precipitation / cold-dissolution cycle was repeated 3–5 times, and an intermediate sample was taken after every step and kept for SDS-PAGE analysis.
9. After the final cold dissolution the solution was filtered through a 0.22 µm filter and dialysed against deionised water at 4 °C with several buffer changes.

Note

The reversible phase transition of the SELP backbone allowed the target protein to be purified without an affinity tag. The intermediate samples collected throughout the purification showed that the contaminating proteins were progressively removed with each transition and that the purity of the 2KY-5 band increased from cycle to cycle; the number of cycles after which the target band was essentially free of contaminants is recorded in the figure below. Because the transition behaviour depended on the salt concentration and temperature, the conditions were refined further in the next set of experiments.

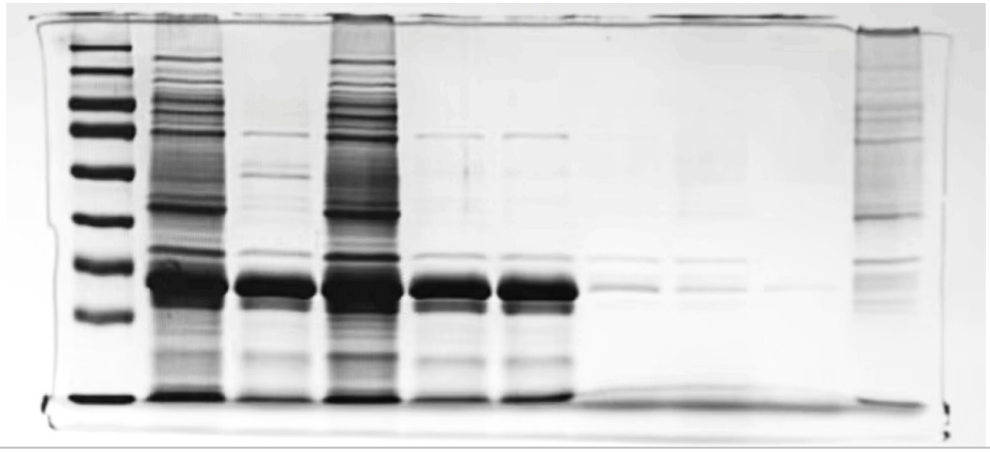


Figure 7. SDS-PAGE of the intermediate samples taken during the ITC purification of 2KY-5 .

4.8 Iterative optimisation of the 2KY-5 purification conditions

Date: 19–21 July and 2–4 August 2026

Procedures

1. The ITC purification of 2KY-5 was repeated with a matrix of salt concentrations (1.0, 1.5, 2.0, 2.5 and 3.0 M NaCl) combined with a range of transition temperatures (25, 30, 37 and 42 °C) to identify the combination that gave the cleanest product with the highest recovery.
2. The lysis, hot-precipitation, centrifugation and cold-dissolution steps were kept identical to record 4.7 so that the effect of the salt and temperature parameters could be compared directly; the onset of turbidity was recorded for each condition.
3. Additional variables were tested in the second round on 2–4 August, including the centrifugation temperature, the length of the cold-dissolution step, the number of transition cycles and the protein concentration of the lysate.

4. Intermediate samples were taken after every step of every condition and analysed by SDS-PAGE; the yield of the target band and the level of contaminating bands were compared between conditions.
5. The best-performing combination was defined as the standard purification condition and was reused without further change for all later 2KY-5 and 2KY-10 batches.

Note

The purity and the recovery of 2KY-5 both depended strongly on the salt concentration and on the temperature of the hot-precipitation step; conditions that were too mild left the target protein in the discarded supernatant, whereas conditions that were too harsh co-precipitated contaminants and reduced the redissolution efficiency. A single optimal combination was identified from the SDS-PAGE comparison and adopted as the standard condition: NaCl concentration 2 M, transition temperature 65 °C. The optimised condition gave a visibly cleaner 2KY-5 band and a higher recovery than the initial protocol and was used for every subsequent batch of scaffold protein.

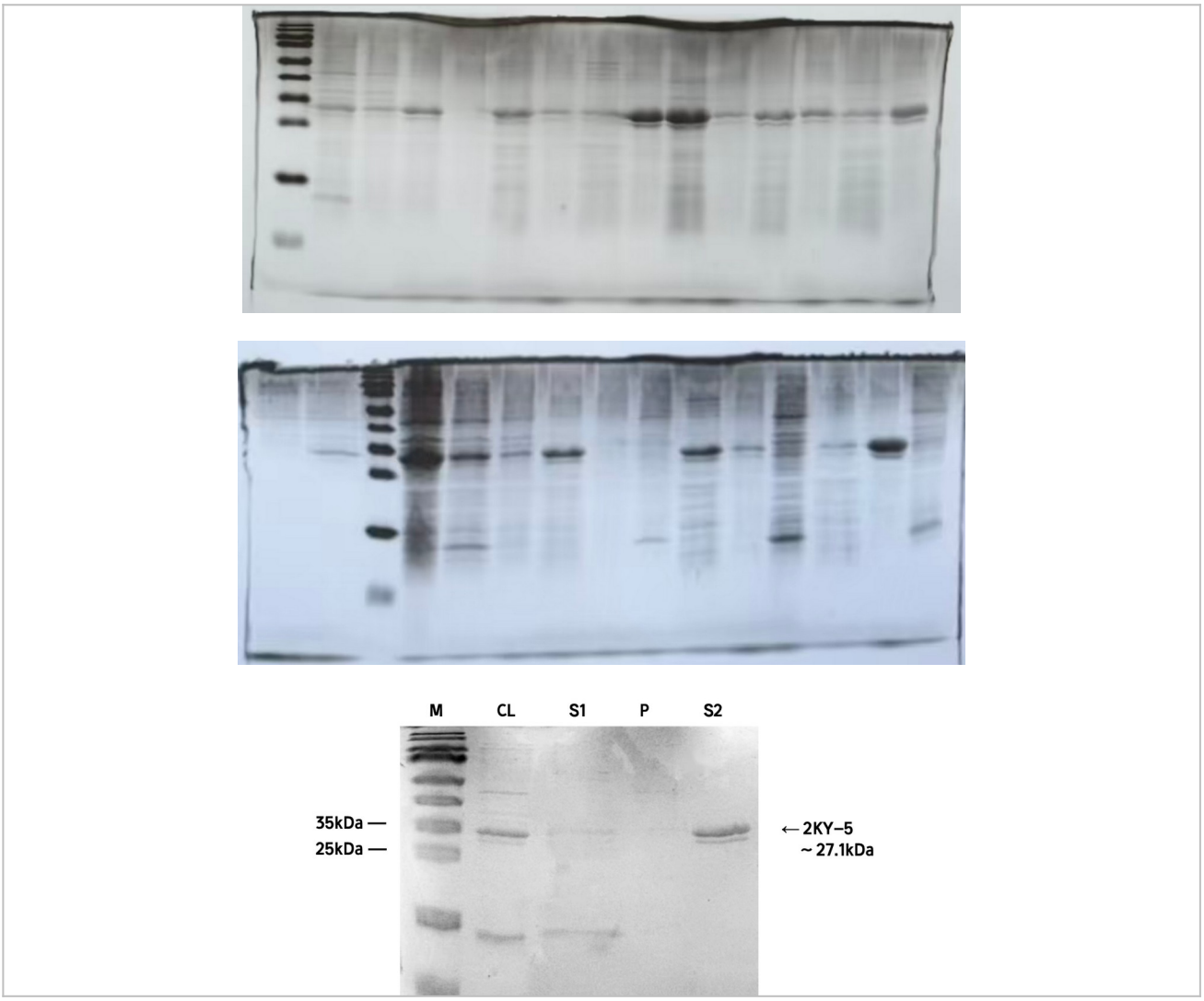


Figure 8. SDS-PAGE comparison of the 2KY-5 purification under the salt and temperature gradients tested on 19–21 July and 2–4

9 Dialysis, desalting and lyophilisation of 2KY-5

Date: 22–24 July 2026

Procedures

1. The purified 2KY-5 solution was loaded into dialysis tubing with an appropriate molecular-weight cut-off, sealed with clips and dialysed against at least 100 volumes of deionised water at 4 °C for 2 days with 3–4 buffer changes to remove the salt completely.
2. The dialysed solution was centrifuged at 4 °C, 12,000 g for 10 min to remove any insoluble material and the supernatant was retained.
3. The protein concentration was measured by A280, and the solution was aliquoted into freeze-drying containers filled to no more than one third of their volume.
4. The samples were pre-frozen at −80 °C for at least 2–4 h until completely solid.
5. The frozen samples were transferred quickly to the freeze-dryer and lyophilised under vacuum for 24–48 h until a white, porous powder of constant weight was obtained.
6. The vacuum was vented slowly, the powder was weighed to calculate the recovery, sealed and stored at 4 °C with desiccant until use.

Note

Dialysis removed the residual salt and the purified 2KY-5 was recovered as a white, porous and easily weighed powder. The recovery per litre of culture is recorded in the figure below. Complete desalting was important, because residual salt both reduced the freeze-drying efficiency and made the powder more difficult to redissolve; the powder dissolved readily in PBS at 4 °C to give the 10% (w/v) precursor solution used in the crosslinking experiments.

10 Large-scale expression of the Ligation Syste proteins

Date: 26–28 July 2026

Procedures

1. Starter cultures of SnoopTag2-sfGFP, SnoopLigase2 and SUMO-DogTag were grown in 10 mL of LB broth with antibiotic at 37 °C, 200 rpm for 16 h.
2. Each starter culture was diluted 1:100 into 1 L of LB broth with antibiotic in baffled flasks and grown at 37 °C, 200 rpm until the OD600 reached 0.6–0.8.
3. Expression was induced with the optimal IPTG concentration determined in the small-scale trials, and the cultures were incubated at the corresponding optimal temperature and duration for each construct.
4. Samples were taken before induction and at the end of induction and normalised by OD600 for SDS-PAGE verification of the large-scale expression.
5. The cells were harvested by centrifugation at 4 °C, 4,000–6,000 g for 15–20 min, washed once with pre-chilled PBS and stored at −80 °C until purification.

Note

The large-scale cultures expressed all three Ligation Syste proteins at the expected molecular weights, and the expression level was comparable to or higher than in the small-scale trials. The cell pellets were stored at −80 °C, which also improved subsequent lysis.

11 Large-scale Ni-NTA purification of the Ligation Syste proteins and dialysis

Date: 29–30 July 2026

Procedures

1. The cell pellets from the large-scale expression were resuspended at 1:10 (w/v) in ice-cold Ni-NTA buffer supplemented with 10 mM imidazole, protease inhibitor cocktail and 1 mM PMSF.
2. The cells were lysed on ice by sonication until the suspension cleared, and the lysate was centrifuged at 4 °C, 30,000–35,000 g for 30–40 min to remove cell debris.
3. The clarified lysate was incubated with Ni-NTA resin (1–2 mL of resin slurry per litre of culture) at 4 °C with gentle inversion for 1 h.
4. The slurry was packed into a gravity column and the flow-through was collected; the resin was washed with 3 × 10 column volumes of buffer containing 10 mM imidazole.
5. The bound protein was eluted with buffer containing 200 mM imidazole, and the fractions were analysed by SDS-PAGE; the purest fractions were pooled.
6. The pooled eluates were concentrated by ultrafiltration with a membrane cut-off chosen to be about one third of the protein molecular weight, and buffer-exchanged into PBS or Tris-borate buffer pH 7.4.
7. The concentrated proteins were dialysed against the storage buffer at 4 °C for 12–24 h with 2–3 buffer changes to remove the imidazole completely; glycerol was added to 15% (v/v) where required for storage.
8. The final protein concentrations were measured by A280, and the proteins were aliquoted and stored at −80 °C.

Note

The large-scale Ni-NTA purification gave the three Ligation Syste proteins in high purity and in sufficient quantity for the ligation reactions. The imidazole was removed by dialysis before the ligation, because residual imidazole and salt interfere with the SnoopLigase2 reaction. The final concentrations and the total yields obtained from each litre of culture are recorded in the figure below.

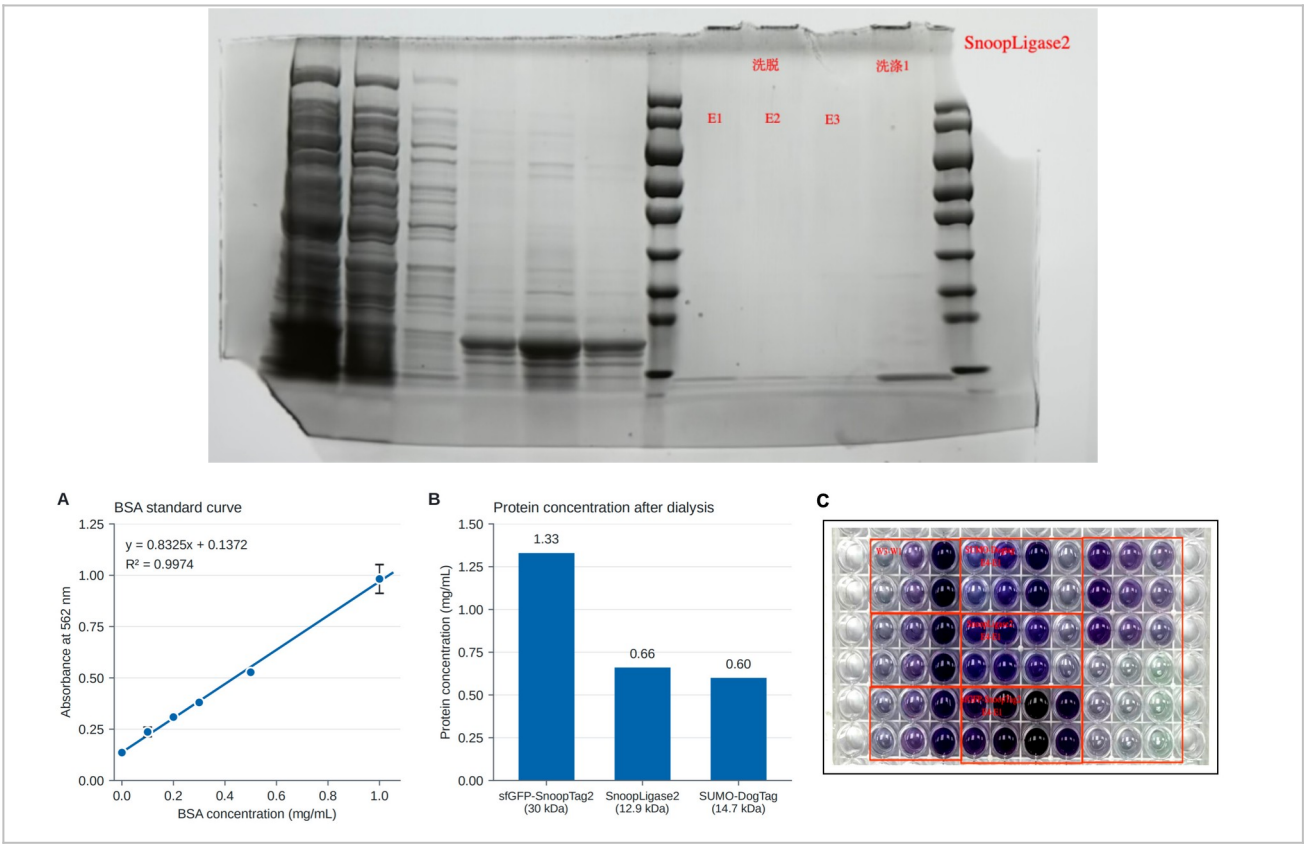


Figure 11. SDS-PAGE of the large-scale Ni-NTA purification of the Ligation Syste proteins (L, lysate; FT, flow-through; W, wash; E1–E4, elution fractions; P, pooled and dialysed protein).

12 Ligation reaction and SDS-PAGE evaluation of the coupling efficiency

Date: 31 July 2026

Procedures

1. SnoopTag2-sfGFP, SUMO-DogTag and SnoopLigase2 were diluted to working concentration in Tris-borate buffer pH 7.4 and kept on ice.
2. The reaction was assembled at equimolar concentration (20–25 µM of each component) in a total volume of 20–100 µL; a negative control without SnoopLigase2 was set up in parallel.
3. The reaction was incubated at 21 °C for 3 h, and a preparative reaction was additionally incubated at 4 °C with gentle rotation for 16 h.
4. Aliquots of 9 µL were withdrawn at 0, 5, 10, 15, 30 and 60 min and at the endpoint, and each was quenched immediately with 6× SDS loading buffer containing 100 mM DTT.
5. The quenched samples were heated at 95 °C for 3 min and separated on a 16% (w/v) SDS-PAGE gel with Coomassie staining.
6. The band intensities were quantified and the coupling efficiency was calculated as 100 × product / (product + unreacted SnoopTag2 component + unreacted DogTag component); the experiment was performed in triplicate.

Note

A new band of the expected size of the ligation product appeared in the complete reaction, while the negative control without SnoopLigase2 showed essentially no product, confirming that the ligation was enzyme-dependent.

The coupling efficiency increased with time and reached ____% after 3 h at 21 °C; the corresponding values for the 4 °C overnight reaction are recorded in the figure below. The reaction conditions of this test were used for the later ligation of the SnoopTag2-tagged scaffolds.

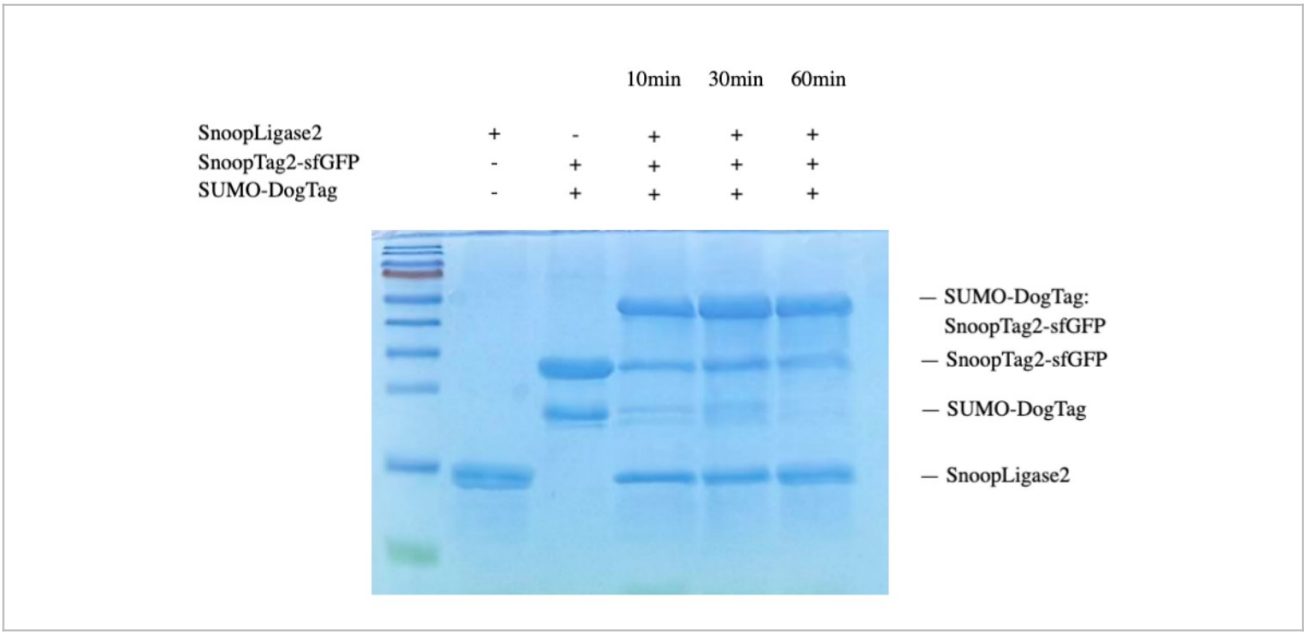


Figure 12. Time course of the SnoopLigase2-mediated ligation analysed by SDS-PAGE

13 Recovery of the 2KY-5 cloning strain and plasmid mini-prep

Date: 8–9 August 2026

Procedures

1. The glycerol stock of the 2KY-5/DH5α cloning strain was streaked onto an LB agar plate with antibiotic and incubated at 37 °C overnight for recovery.
2. A single colony was inoculated into 5 mL of LB broth with antibiotic and grown at 37 °C, 200 rpm for about 16 h.
3. The cells were harvested by centrifugation at 4 °C, 4,000–6,000 g for 10–15 min, and the plasmid DNA was extracted by alkaline lysis followed by binding to a silica column according to the manufacturer's instructions.
4. The column was washed twice with ethanol-containing wash buffer, dried, and the plasmid was eluted with 30 –50 µL of elution buffer or nuclease-free water.
5. The plasmid concentration and the A260/A280 ratio were measured with a Nanodrop, and 2–5 µL of the preparation was checked by agarose gel electrophoresis.
6. The plasmid DNA was stored at −20 °C.

Note

The plasmid was obtained, but the concentration measured by Nanodrop was considerably lower than expected for a routine mini-prep, and the amount of DNA was not sufficient for the following cloning work. Because the

backbone had never been propagated at this scale in our hands, the low yield was attributed to the combination of a low-copy behaviour of the construct and the standard extraction conditions, and a troubleshooting round was started

14 Troubleshooting the low plasmid yield of 2KY-5

Date: 10–13 August 2026

Procedures

1. The plasmid preparation was repeated in parallel with a concentration step: several elution fractions were pooled and concentrated by isopropanol/ethanol precipitation or by vacuum concentration to raise the final DNA concentration.
2. The culture conditions were modified: a larger culture volume was used, the culture was grown at 37 °C, 200 rpm only until the mid-exponential phase (OD600 0.3–0.5), and chloramphenicol was then added to 170 µg/mL before the culture was continued at 37 °C, 200 rpm for a further 12–16 h to amplify the relaxed plasmid while host protein synthesis and chromosome replication were inhibited.
3. The plasmid extraction conditions were adjusted in parallel: the resuspension volume of Solution I was increased and the pellet was resuspended completely to avoid clumps, the lysis time was controlled carefully, and the elution buffer was pre-warmed and incubated on the column for 5–10 min before the first elution (heated elution incubation).
4. The elution volume was reduced to 30 µL and the column was eluted a second time to recover the remaining DNA; the two eluates were compared by Nanodrop and agarose gel electrophoresis.
5. The combinations of culture and extraction conditions were compared side by side, and the parameters giving the highest yield and the best A260/A280 ratio were fixed as the standard procedure for this construct.

Note

The combination of chloramphenicol amplification at the mid-exponential phase and the heated elution incubation produced a clear improvement in the plasmid yield, and the A260/A280 ratio remained within the accepted range. The optimised parameters were written into our working procedure and were used for every subsequent preparation of the 2KY-5 and 2KY-10 plasmids. The yield before and after optimisation is compared in the figure below (____ ng/µL before, ____ ng/µL after).

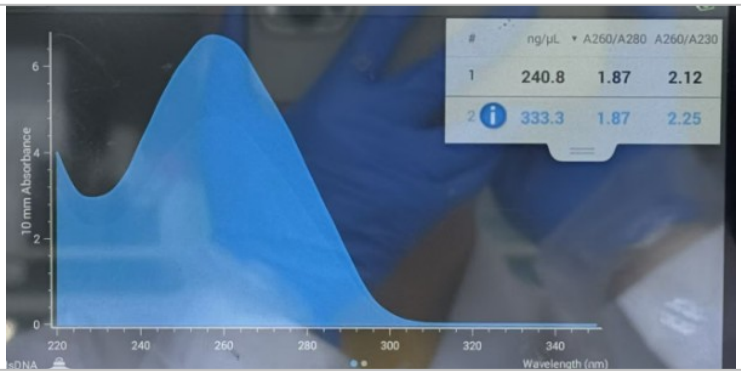


Figure 14. Nanodrop concentrations of the different culture and extraction conditions tested.

15 Construction of 2KY-10 by recursive directional ligation (PRe-RDL)

Date: 16–19 August 2026

Procedures

1. The 2KY-5 plasmid obtained with the optimised procedure was used as the parental template for the construction of the higher-repeat variant 2KY-10.
2. A recursive directional ligation by plasmid reconstruction (PRe-RDL) strategy was applied: the repeat-unit coding sequence was amplified with primers carrying the appropriate terminal sequences, and the backbone and the insert were designed so that they could be joined in a defined orientation in a single seamless assembly reaction.
3. The PCR products were purified, and the concentration of each fragment was measured to set up the assembly reaction at the calculated molar ratio.
4. The assembly reaction was incubated at the recommended temperature, then used to transform competent DH5α cells by heat shock.
5. The transformation was plated on selective LB agar and incubated at 37 °C overnight.
6. Single colonies were screened by colony PCR with a primer pair spanning the junction between the vector and the repeated insert, so that a clone carrying the doubled repeat could be distinguished from the parental plasmid.
7. Positive clones were grown overnight, the plasmid was prepared and the repeat number and the sequence of the junction were verified by restriction analysis and Sanger sequencing.

Note

Colony PCR identified clones whose amplicon was larger than that of the parental 2KY-5 plasmid by the size of one repeat unit, and the sequence of the junction was confirmed by Sanger sequencing. The resulting plasmid was named 2KY-10 and corresponds to a doubling of the repeat number of 2KY-5, doubling the molecular weight of the encoded backbone protein; it was used for all of the subsequent high-molecular-weight hydrogel experiments. The colony PCR gel of the screened clones is shown in the figure below.

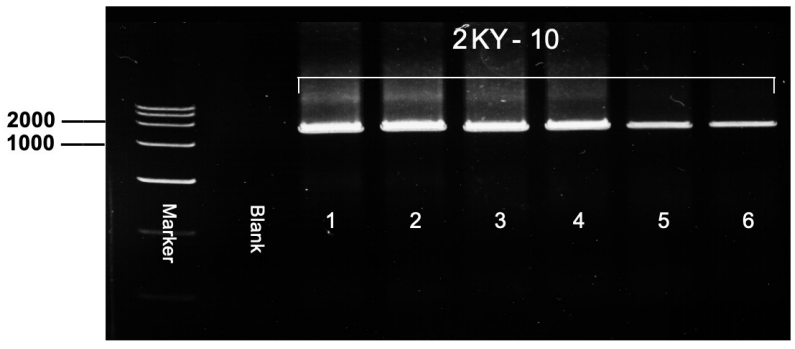


Figure 15. Colony PCR screening of the DH5α clones obtained after the PRe-RDL assembly

16 Plasmid mini-prep of 2KY-10 and transformation into BL21(DE3)

Date: 20–21 August 2026

Procedures

1. A positive 2KY-10/DH5α clone from the PRe-RDL screening was inoculated into LB broth with antibiotic and grown overnight at 37 °C, 200 rpm.
2. The plasmid was extracted with the optimised procedure established in record 4.14, including chloramphenicol amplification and the heated elution incubation.
3. The concentration and purity of the plasmid were checked by Nanodrop and agarose gel electrophoresis, and 1 –2 µL (about 50–100 ng) was used for the transformation.
4. Competent BL21(DE3) cells were thawed on ice, mixed gently with the plasmid, kept on ice for 30 min, heat-shocked at 42 °C for 45 s and returned to ice for 2 min.
5. After recovery in antibiotic-free LB broth at 37 °C, 200 rpm for 45–60 min, the cells were spread onto selective LB agar and incubated at 37 °C for 12–16 h.
6. Single colonies were picked for the expression experiments and for the preparation of glycerol stocks.

Note

The optimised plasmid preparation gave a 2KY-10 plasmid concentration comparable to that of 2KY-5, and the transformation of BL21(DE3) yielded well-separated colonies on the selective plates. Glycerol stocks of the 2KY-10/BL21(DE3) expression strain were prepared and stored at −80 °C, and the plates were used directly for the expression experiment on the following day.

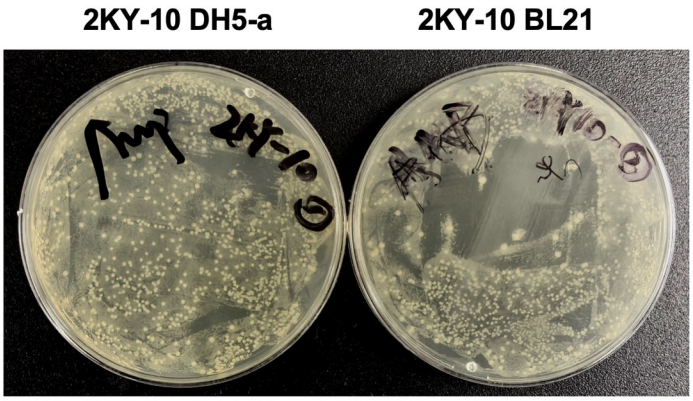


Figure 16. transformed product of 2KY-10

17 Expression of 2KY-10 and SDS-PAGE verification

Date: 22–25 August 2026

Procedures

1. A single colony of the 2KY-10/BL21(DE3) strain was inoculated into 10 mL of LB broth with antibiotic and grown at 37 °C, 200 rpm for 16 h.
2. The starter culture was diluted 1:100 into fresh LB broth with antibiotic and grown at 37 °C, 200 rpm until the OD600 reached 0.5–0.8.

3. Expression was induced with the optimal IPTG concentration determined for the backbone protein, and the culture was incubated at the low induction temperature identified for 2KY-5 in record 4.3, which favours soluble expression of the SELP constructs.
4. Samples were taken before and after induction, normalised by OD600, pelleted and resuspended in 1× SDS loading buffer.
5. The samples were heated at 95 °C for 5–10 min (whole-cell boiling lysis) and separated on an 8–12% (w/v) SDS-PAGE gel suited to the higher molecular weight of 2KY-10, with a pre-stained marker and the 2KY-5 sample loaded alongside for comparison.
6. The gel was stained with Coomassie, destained, imaged and the band positions and intensities were compared.

Note

The induced 2KY-10 sample showed a clear band at a molecular weight approximately twice that of the 2KY-5 band, confirming that the doubled-repeat construct was expressed as a full-length protein and that induction was successful. Minor lower-molecular-weight bands were also observed and were attributed to partial translation products or to the intrinsic heterogeneity of the repetitive sequence; these were removed in the following purification step.

18 ITC purification, dialysis and lyophilisation of 2KY-10

Date: 26–29 August 2026

Procedures

1. The 2KY-10 cell pellet was resuspended in ice-cold PBS with protease inhibitors, lysed on ice and centrifuged to obtain the clarified lysate.
2. The ITC purification was carried out with the optimised salt concentration, transition temperature and number of cycles established in record 4.8, with an intermediate sample taken after every step for SDS-PAGE.
3. The purified 2KY-10 solution was dialysed against deionised water at 4 °C for 2 days with several buffer changes to remove the salt.
4. The dialysed solution was centrifuged to remove insoluble material, the concentration was measured by A280 and the solution was aliquoted into freeze-drying containers.
5. The samples were pre-frozen at −80 °C and lyophilised under vacuum for 24–48 h until a constant-weight white powder was obtained.
6. The powder was weighed, sealed and stored at 4 °C with desiccant for the crosslinking experiments.

Note

The optimised ITC condition transferred from 2KY-5 worked equally well for the higher-molecular-weight 2KY-10 and gave a protein band essentially free of contaminants, confirming that the parameters established in record 4.8 are suitable for both repeat lengths. 2KY-10 was recovered as a white powder, and the powder was used together with the 2KY-5 powder in the photo-crosslinking experiments described in records 19 and 20.

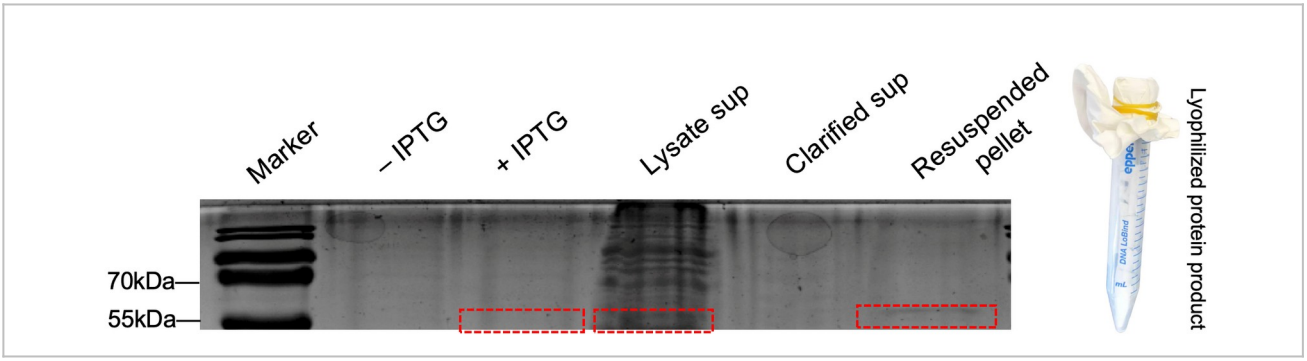


Figure 18. SDS-PAGE of the intermediate samples of the 2KY-10 ITC purification and a photograph of the lyophilised 2KY-10 powder.

19 Preparation of the Ru/SPS photo-crosslinking system (2/20 mM) and blue-light crosslinking

Date: 1 September 2026

Procedures

1. The lyophilised scaffold protein powder (2KY-5 and 2KY-10) was dissolved in PBS at 4 °C with slow stirring to give a 10% (w/v) precursor solution; the solution was centrifuged at 4 °C, 12,000 g for 10 min to remove insoluble material.
2. Ru and SPS stock solutions were prepared fresh, protected from light, and added to the precursor solution at a Ru:SPS molar ratio of 1:10 to give final concentrations of 2 mM Ru and 20 mM SPS (2/20 mM).
3. The mixture was mixed gently by pipetting, avoiding the introduction of bubbles, and kept on ice and protected from light until use.
4. Aliquots of 100–200 µL of the precursor/initiator mixture were transferred into a mould or into the wells of a multi-well plate.
5. The samples were irradiated with a blue LED light source at 450 nm, keeping the irradiation distance and intensity constant between samples.
6. Gelation was confirmed by the inversion test: the container was inverted for 10 s and the sample was scored as crosslinked when it did not flow.
7. The hydrogels were washed three times with PBS to remove unreacted initiator residues and were then used for the following experiments.

Note

The precursor solution containing 2/20 mM Ru/SPS formed a transparent, self-supporting hydrogel under blue light irradiation; the inversion test was passed, and the hydrogel retained its shape during handling and washing. Gelation was visibly faster than in the initial attempts, and the 2/20 mM formulation was taken forward as the standard photo-initiator concentration for the following experiments, in agreement with the concentration reported to give the best balance of crosslinking speed, mechanical properties and cytocompatibility. Photographs of the hydrogel and of the inversion test are shown in the figure below.

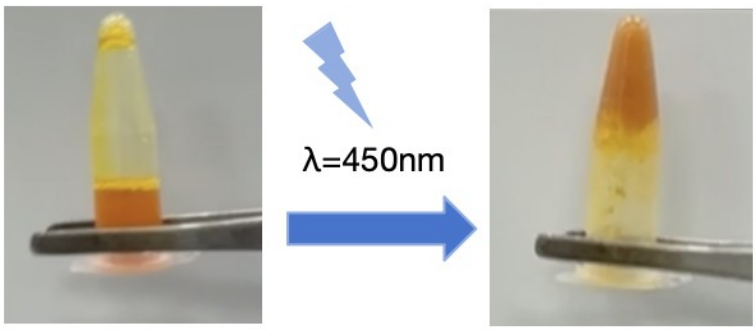


Figure 19. Photographs of the SELP precursor solution, of the blue-light irradiation set-up and of the resulting hydrogel

20 Re-optimisation of the blue-light crosslinking parameters

Date: 3 September 2026

Procedures

1. The photo-crosslinking of the hydrogel backbone was repeated with the same 10% (w/v) precursor solution and the same 2/20 mM Ru/SPS formulation so that only the irradiation parameters were varied.
2. The blue-light wavelength was adjusted, and the effect of the wavelength together with the irradiation time was compared between samples while the distance between the light source and the sample was kept constant.
3. Every sample was evaluated by the inversion test and by visual inspection of transparency and homogeneity, and the handling behaviour of the resulting hydrogels was compared.
4. The combination of wavelength, irradiance and irradiation time that gave complete and homogeneous crosslinking was selected and fixed as the standard condition for the following experiments.

Note

Adjusting the blue-light wavelength parameters gave a clearly improved crosslinking result: the hydrogel became fully solid within the irradiation period, with a homogeneous, transparent appearance and no liquid residue in the mould, whereas the earlier conditions had left softer and less uniform samples. The optimised irradiation parameters were adopted as the standard condition and were used for the preparation of all hydrogels in the cytotoxicity and cell-adhesion experiments.

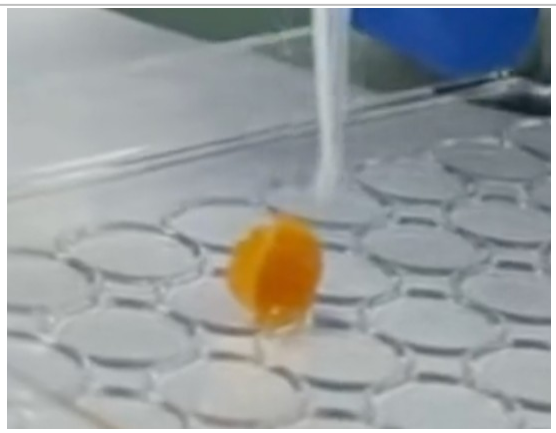


Figure 20. hydrogels obtained after the adjustment of the blue-light parameters

21 Production and purification of SnoopTag2-2KY-5-SnoopTag2 and SnoopTag2-2KY-10-SnoopTag2

Date: 4–8 September 2026

Procedures

1. The glycerol stocks of the strains carrying the SnoopTag2-2KY-5-SnoopTag2 and SnoopTag2-2KY-10-SnoopTag2 constructs were recovered on selective LB agar plates at 37 °C overnight.
2. The plasmids were transformed into competent BL21(DE3) cells by heat shock, plated on selective LB agar and incubated at 37 °C for 12–16 h; the colonies were verified by colony PCR and the constructs were confirmed by sequencing.
3. Starter cultures were grown overnight and diluted 1:100 into fresh LB broth with antibiotic; the cultures were grown to mid-exponential phase and induced with the optimal IPTG concentration using the low-temperature induction condition established for the SELP backbone.
4. Samples were taken before and after induction for SDS-PAGE verification, and the cells were harvested by centrifugation at 4 °C and stored at −80 °C.
5. Both tagged scaffold proteins were purified by the optimised ITC procedure with intermediate samples taken after every step for SDS-PAGE analysis.
6. The purified proteins were filtered, dialysed against deionised water or the ligation buffer at 4 °C and, where required, lyophilised and stored at 4 °C.
7. The identity and purity of the two tagged scaffolds were confirmed by SDS-PAGE, where the SnoopTag2 fusion appeared at a molecular weight higher than that of the corresponding untagged backbone.

Note

Both SnoopTag2-2KY-5-SnoopTag2 and SnoopTag2-2KY-10-SnoopTag2 were successfully produced and purified with the procedures established for the untagged backbone, and the tagged constructs migrated at the expectedly higher molecular weights. The tagged scaffolds were obtained in sufficient quantity for the ligation reactions and were shown to retain the reversible phase-transition behaviour required by the ITC purification.

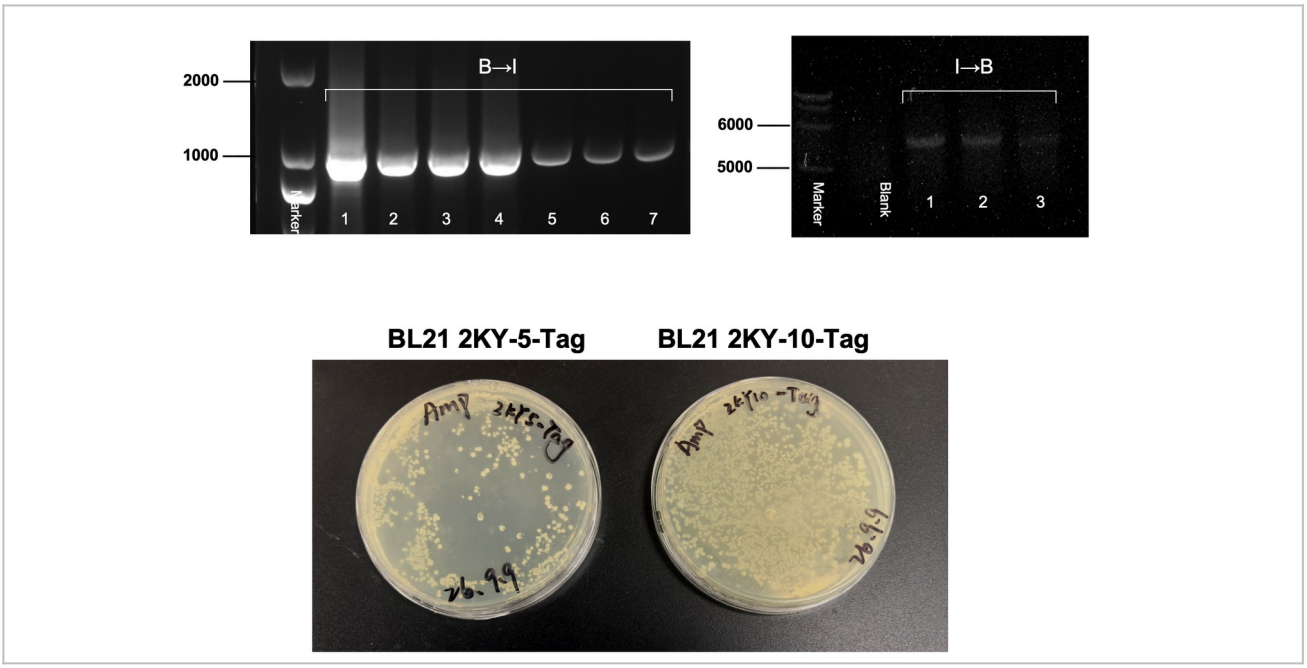


Figure 21. trandformed product of 2KY-5-Tag and 2KY-10-Tag

22 Ligation of the tagged scaffolds with SUMO-DogTag and SnoopLigase2

Date: 10 September 2026

Procedures

1. SnoopTag2-2KY-5-SnoopTag2 and SnoopTag2-2KY-10-SnoopTag2 were diluted together with SUMO-DogTag and SnoopLigase2 in Tris-borate buffer pH 7.4 and kept on ice.
2. The ligation reactions were assembled with the tagged scaffold, SUMO-DogTag and SnoopLigase2 at the molar ratio established in record 4.12, and negative controls without SnoopLigase2 were prepared in parallel.
3. The reactions were incubated at 21 °C for 3 h, with an additional reaction incubated at 4 °C with gentle rotation overnight.
4. Aliquots were withdrawn at defined time points and quenched immediately with 6× SDS loading buffer containing DTT.
5. The quenched samples were heated at 95 °C for 3 min and analysed by SDS-PAGE with Coomassie staining, using a gel percentage appropriate for the high molecular weight of the scaffolds.
6. The band pattern was compared with that of the untreated scaffold and the extent of the reaction was estimated from the band intensities.

Note

For both tagged scaffolds a new high-molecular-weight band appeared after the reaction, corresponding to the scaffold carrying the ligated SUMO-DogTag partner, and the intensity of this band increased with time; the negative controls without SnoopLigase2 showed no such band. The results confirm that the SnoopTag2 sequences fused to both termini of the SELP backbone remain accessible and reactive and that the hydrogels can be covalently functionalised through the SnoopLigase2 reaction.

23 Cytotoxicity of the hydrogel extracts evaluated by CCK-8

Date: 12–14 September 2026

Procedures

1. Hydrogels were prepared from the 10% (w/v) precursor solution with the standard 2/20 mM Ru/SPS formulation and crosslinked under the optimised blue-light irradiation condition established in record 4.20.
2. The hydrogels were sterilised and incubated in complete culture medium at 37 °C for 24 h to obtain the hydrogel extract.
3. The extract was collected and filter-sterilised, and cells were seeded in 96-well plates and cultured overnight to allow attachment.
4. The culture medium was replaced with the hydrogel extract, while cells cultured in complete medium served as the blank control.
5. After 1 and 3 days of culture, the medium was removed, the cells were washed with PBS and incubated with culture medium containing 10% (v/v) CCK-8 solution at 37 °C for 2 h.
6. The optical density was measured at 450 nm with a microplate reader, and the cell viability of the extract group was expressed relative to the blank control; the experiment was performed in triplicate.

Note

The cells cultured in the hydrogel extract remained viable and proliferated over the 3-day period, and the OD450 values of the extract group were comparable to those of the blank control, indicating that the extract of the crosslinked hydrogel caused no significant cytotoxicity. The results confirm that the 2/20 mM Ru/SPS crosslinking system and the residual components of the washed hydrogel are compatible with the cells used in this study.

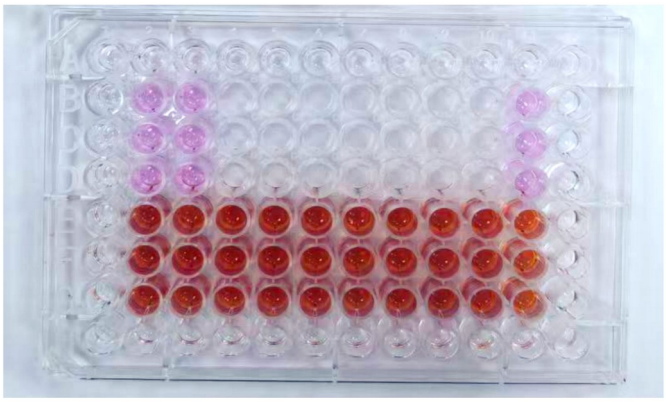


Figure 23. CCK-8 cell-viability assay of the hydrogel extract, normalised to the blank control.

4.24 Cell adhesion on the hydrogel

Date: 19–21 September 2026

Procedures

1. Hydrogels were prepared in 48-well plates with the standard formulation and crosslinking condition and sterilised under ultraviolet light.

2. A 200 µL aliquot of cell suspension (2 × 10^4 cells/mL in complete medium) was seeded onto the surface of each hydrogel.
3. The constructs were incubated at 37 °C in a humidified atmosphere with 5% CO_2 for 4 h for the short-term adhesion test, and a parallel set was cultured for 1 and 3 days to follow longer-term adhesion and proliferation.
4. At each time point the culture medium was removed and the hydrogels were washed gently with PBS.
5. The attached cells were stained with a live/dead assay kit (Calcein-AM and propidium iodide in PBS) for 15 min at 37 °C in the dark.
6. The samples were imaged by fluorescence or confocal laser scanning microscopy, and the number of attached cells and their morphology were analysed with image-analysis software.

Note

Cells attached to the surface of the hydrogel within the 4 h incubation and remained attached and viable over the following days; the live/dead staining showed predominantly viable cells, and part of the population displayed a spread morphology rather than a rounded shape, which indicates that the RGD motifs encoded in the SELP backbone are functional and support cell adhesion.

References

[1] Nie K, et al. Ultrafast crosslinking, strongly adhesive de novo protein hydrogels promote cartilage regeneration. Bioactive Materials, 2026, 56: 368–385. doi:10.1016/j.bioactmat.2025.10.009. (SELP design, ITC purification, Ru/SPS blue-light crosslinking, CCK-8 and cell adhesion methods)

[2] Keeble AH, Wood DP, Howarth M. Design and Evolution of Enhanced Peptide–Peptide Ligation for Modular Transglutaminase Assembly. Bioconjugate Chemistry, 2023, 34(10): 1019–1036. doi:10.1021/acs.bioconjchem.3c00122. (SnoopTag2 / DogTag2 / SnoopLigase2 ligation system)

[3] McDaniel JR, Mackay JA, Quiroz FG, Chilkoti A. Recursive directional ligation by plasmid reconstruction allows rapid and seamless cloning of oligomeric genes. Biomacromolecules, 2010, 11(4): 944–952. doi:10.1021/bm901387t. (PRe-RDL method used to construct 2KY-10)

[4] iGEM Team laboratory protocols P01–P14 (companion documents): plasmid transformation, IPTG-induced expression, SDS-PAGE, Western blot, ultrafiltration, ITC purification, lyophilisation, hydrogel photo-crosslinking, Ni-NTA purification, dialysis, ligation reaction, low-copy plasmid preparation, colony PCR and Gibson assembly.